Magnetic matrix statements in Clean

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LLM-assisted specimen. Source attributions retain their original authorship; see PROVENANCE.md.

These six statements have Lean proofs in `studies/magnetic/Verified.lean`. The Laplacian uses the unnormalized convention $L_q = D - M_q$, with $D_{ii} = \sum_j (A_{ij} + A_{ji})$. The inverse endpoint result is a counterexample to the displayed odd-length specialization of v1 equation (24). The matrix E_{01} is $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$.

Result 1 (magnetic powers encode walk profiles). [Huang, Gleich, Li 2025, equation (1), Supplement S.II equation (17), Theorem 1]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{R}$. Let m be a natural number. Let q be a real number. Then

$$M_q(A)^m = e^{2\pi i - (q\,m)}\ \mathrm{sumBefore}(m+1, \Big(k \mapsto e^{2\pi i 2\,q\,k}\ \Phi_A(m,k)\Big)).$$

Adapted from the cited source. Scope: forward transform only.

Result 2 (magnetic adjacency is Hermitian). [Huang, Gleich, Li 2025, magnetic graph matrix definition]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{R}$. Let q be a real number. Then

$$M_q(A) \text{ is Hermitian.}$$

Adapted from the cited source.

Result 3 (magnetic Laplacian is Hermitian). [Clean magnetic formalization, unnormalized D minus M convention]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{R}$. Let q be a real number. Then

$$L_q(A) \text{ is Hermitian.}$$

Result 4 (zero potential symmetrizes adjacency). [Huang, Gleich, Li 2025, discussion following equation (1)]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{R}$. Then

$$M_0(A) = A + A^{\mathsf{T}}.$$

Adapted from the cited source. Scope: first power.

Result 5 (two step mixed walk profile). [Huang, Gleich, Li 2025, walk profile examples preceding equation (1)]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{R}$. Then

$$\Phi_A(2,1) = A\,A^{\mathsf{T}} + A^{\mathsf{T}}\,A.$$

Adapted from the cited source.

Result 6 (odd length inverse endpoint counterexample). [Clean magnetic formalization, inverse_v1_length_one_fails]
Then

$$\Phi_{E_{01}}(1,1) \neq \frac{1}{2}\,M_0(E_{01}).$$

Scope: v1 equation (24) at m = 1.

References

Huang, Gleich, Li 2025 Yinan Huang, David F. Gleich, Pan Li. Powers of Magnetic Graph Matrix: Fourier Spectrum, Walk Compression, and Applications. Version arXiv:2506.07343v1. Source

Clean magnetic formalization . Supporting results for the magnetic graph matrix.